

Enroute

Full-Stack Logistics Platform

A platform connecting truck drivers (with empty return trips) to individuals and small businesses who need affordable courier/transport services.

Tech Stack

Frontend

Layer	Technology	Why
UI Framework	React 18 (Vite)	Fast dev, component-based UI
Routing	React Router v6	SPA page navigation
Styling	Tailwind CSS	Rapid UI prototyping
HTTP Client	Axios	API calls to FastAPI backend
State Mgmt	React Context	Lightweight, built-in
Notifications	React Hot Toast	Toast messages
Maps	React-Leaflet + OSM	Free, no API key, interactive maps

Backend (Python)

Layer	Technology	Why
Framework	FastAPI	Modern, async, auto Swagger docs at /docs
Server	Uvicorn	ASGI server to run FastAPI
ORM	SQLAlchemy 2.0	Python classes -> DB tables
Migrations	Alembic	Auto DB schema migrations
Validation	Pydantic (built-in)	Request/response validation, zero setup
Auth	python-jose + passlib	JWT tokens + password hashing
CORS	CORSMiddleware	Built-in, let React talk to FastAPI
Environment	python-dotenv	Secrets in .env
DB Driver	psycopg2-binary	PostgreSQL adapter for SQLAlchemy

Database

Supabase (Hosted PostgreSQL) - Free managed PostgreSQL with dashboard. Grab the connection string and plug into SQLAlchemy.

Deployment

Component	Platform	Why
Frontend	Vercel	One-click React deploys, auto on git push, free
Backend	Render.com (free)	Proper FastAPI server, GitHub auto-deploy
Database	Supabase	Free hosted PostgreSQL, web dashboard, 500MB
Source Control	GitHub	Mono-repo with client/ and server/ folders

Why not Vercel for the backend? Vercel converts Python apps to serverless functions, which causes issues with FastAPI persistent DB connections. Render.com gives you a real server with zero config.

Architecture (Deployed)

```
React App (Vercel) <-- HTTPS / Axios --> FastAPI Backend (Render.com)
enroute.vercel.app                      enrout-api.onrender.com
                                         |
                                         SQLAlchemy
                                         (connection string)
                                         |
                                         Supabase PostgreSQL
                                         (Free Tier)
```

Proposed Folder Structure

Frontend (client/)

```
client/
  public/
    enrout-logo.svg
  src/
    assets/
    components/
      Navbar.jsx, Footer.jsx, RouteCard.jsx, SearchBar.jsx,
      FilterPanel.jsx, BookingModal.jsx, RouteMap.jsx, ProtectedRoute.jsx
    pages/
      Home.jsx, Login.jsx, Register.jsx, DriverDashboard.jsx,
      CreateListing.jsx, BrowseRoutes.jsx, RouteDetails.jsx,
      MyBookings.jsx, Profile.jsx
    context/
      AuthContext.jsx
    services/
      api.js, authService.js, routeService.js, bookingService.js
    utils/
      formatDate.js
    App.jsx, main.jsx, index.css
  index.html, tailwind.config.js, vite.config.js, vercel.json, package.json
```

Backend (server/)

```
server/  
  app/  
    __init__.py, main.py, config.py, database.py  
  models/  
    __init__.py, user.py, route_listing.py, booking.py  
  schemas/  
    __init__.py, user.py, listing.py, booking.py  
  routers/  
    __init__.py, auth.py, listings.py, bookings.py  
  dependencies/  
    __init__.py, database.py, auth.py  
  utils/  
    __init__.py, security.py, helpers.py  
alembic/  
  versions/, env.py, alembic.ini  
requirements.txt, render.yaml, .env.example
```

Root Files

.gitignore, LICENSE, README.md

Key Backend Files Explained

File	Purpose
main.py	Creates the FastAPI app, adds CORS, includes all routers
config.py	Uses Pydantic BaseSettings to load .env variables
database.py	SQLAlchemy engine + session factory + Base class
models/*.py	SQLAlchemy table definitions (ORM)
schemas/*.py	Pydantic models for request validation & response
routers/*.py	API endpoint functions (like Flask blueprints)
dependencies/*.py	FastAPI Depends() - inject DB sessions, auth checks
utils/security.py	Password hashing (bcrypt) + JWT create/verify

Database Schema

User

Field	Type	Notes
id	Integer (PK)	Auto-increment
name	String(100)	Full name

Field	Type	Notes
email	String(120)	Unique
password_hash	String(256)	bcrypt
phone	String(15)	Contact
role	Enum	DRIVER / CUSTOMER
created_at	DateTime	Auto-generated

RouteListing

Field	Type	Notes
id	Integer (PK)	
driver_id	FK -> User	Who posted it
origin	String(200)	Starting location
destination	String(200)	Ending location
origin_lat / origin_lng	Float	For Leaflet map markers
dest_lat / dest_lng	Float	For Leaflet map markers
departure_date	DateTime	When truck leaves
distance_km	Float	Distance in km
truck_type	String(50)	e.g. Open Body, Container
truck_capacity	String(50)	e.g. 5 tons
available_space	String(100)	How much space is free
rate_per_km	Float	Price per km
flat_rate	Float (nullable)	Or flat price for trip
description	Text (nullable)	Additional notes
status	Enum	ACTIVE / COMPLETED / CANCELLED
contact_phone	String(15)	Driver's contact
created_at	DateTime	

Booking

Field	Type	Notes
id	Integer (PK)	
customer_id	FK -> User	Who booked
route_id	FK -> RouteListing	Which listing
pickup_location	String(200)	Where to pick up goods
drop_location	String(200)	Where to deliver

Field	Type	Notes
goods_description	String(500)	What's being shipped
estimated_weight	String(50)	Approximate weight
status	Enum	PENDING / CONFIRMED / REJECTED / COMPLETED
created_at	DateTime	

Core API Endpoints

FastAPI auto-generates interactive docs at /docs (Swagger UI) and /redoc (ReDoc). No extra setup needed - every endpoint below will be testable in-browser.

Auth - /api/auth

Method	Endpoint	Description
POST	/register	Register (driver or customer)
POST	/login	Login -> returns JWT access token
GET	/me	Get current user (requires token)

Route Listings - /api/routes

Method	Endpoint	Description
POST	/	Create listing (driver only)
GET	/	Browse + search/filter
GET	/[{id}]	Single route details
PUT	/[{id}]	Update listing (owner only)
DELETE	/[{id}]	Cancel listing (owner only)
GET	/my-listings	Driver's own listings

Bookings - /api/bookings

Method	Endpoint	Description
POST	/	Book a route (customer only)
GET	/my-bookings	Customer's bookings
GET	/driver-requests	Driver's incoming requests
PATCH	/[{id}]/status	Driver accepts/rejects a booking

Design Decisions

#	Question	Decision	Notes
1	Payments	Offline only	No payment gateway. Cash/UPI offline.
2	Map Integration	Leaflet + OSM	Free, no API key. Interactive map.
3	Real-time Chat	Phone + WhatsApp	wa.me/<number> button. No chat infra.
4	Admin Panel	Skip	Out of scope for hackathon.

Verification Plan

Manual Verification:

- Full user flow: Register -> Create Listing -> Browse -> Book -> Confirm
- Test FastAPI auto-docs at /docs
- Test on live URLs (Vercel + Render)
- Mobile responsive check
- Show Supabase dashboard with live data during demo